

Enhanced Differential Gene Expression Analysis of Mesial Temporal Lobe Epilepsy in Neocortex Using Synthetic Controls

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Abstract

This exploratory RNA-seq study investigates challenges in differential gene expression (DGE) analysis for underpowered RNA-seq datasets, using mesial temporal lobe epilepsy (MTLE) as a case study. We analyze neocortex RNA-seq data from GSE-134697, with 17 MTLE and 2 control samples. To assess low sample size effects, we generated three synthetic control replicas using a negative binomial simulation, using mean and dispersion from real controls. Using DESeq2, we compared real-only and synthetic-augmented datasets. Significant DEGs increased from 33 (real only) to 1863 (synthetic augmented), showing how limited replication constrains discovery. GO enrichment revealed synaptic signaling (GO:0007268) as a top term, with genes such as SNAP25, SYT1, and GRIN2B significantly upregulated. However, GO enrichment interpretations must be viewed cautiously as a result of synthetic enhancement that can inflate pathway coverage. This work focuses on methodology, not novel MTLE insights, highlighting the role of synthetic data in educational and exploratory RNA-seq studies.

1 Introduction

Mesial temporal lobe epilepsy (MTLE) is a focal epilepsy subtype characterized by seizures that originate in the temporal lobe, often linked to hippocampal sclerosis and altered cortical excitability. RNA sequencing (RNA-seq) enables transcriptomic analysis, but limited control samples reduce statistical power. This study uses the GSE-134697 dataset, with 17 MTLE and 2 control neocortex samples. We generated three synthetic control samples via negative binomial modeling to explore statistical power and DEG detection, discussing implications for transcriptomic studies.

2 Methods

2.1 Data Acquisition

RNA sequence count matrices and metadata were obtained from GSE-134697 [GEO, 2019]. Neocortex samples (17 MTLE, 2 controls) were retained. Genes with total raw counts ≤ 10 across all samples were excluded.

2.2 Synthetic Control Generation

Three synthetic control samples were simulated using the negative binomial distribution. The mean and dispersion parameters were estimated from the two real controls using DESeq2 [Love et al., 2014], assuming that the gene counts follow this model.

2.3 Differential Expression Analysis

Differential gene expression (DGE) analysis was performed using DESeq2. Two analyses were conducted:

1. Real-only: 17 MTLE vs. 2 controls
2. Synthetic-augmented: 17 MTLE vs. 5 controls (2 real + 3 synthetic)

Genes with adjusted p-value (Benjamini-Hochberg) ≤ 0.05 and $|\log_2 \text{FC}| > 1$ were considered significant.

2.4 Gene Ontology Enrichment

The clusterProfiler package [Yu et al., 2012] was used for Gene Ontology (GO) enrichment on DEG sets. The dot plots visualized the enriched biological processes, with additional visualizations in Supplementary Information.

2.5 Visualization

The plots included PCA, MA plots, volcano plots, and GO dotplots for both analyses, with heatmaps and GO network plots in Supplementary Information. All code and output are available at https://github.com/zeinmursal/mTLE_DGE_Analysis/.

3 Results

3.1 DEG Counts

The real-only analysis identified 33 DEGs (26 up-regulated, 7 down-regulated), while the synthetic-augmented analysis identified 1863 DEGs (1510 up-regulated, 353 down-regulated),

highlighting the impact of sample size on statistical power (Table 1, Figure 1).

Table 1: Significant DEGs Comparison

Analysis	Controls	DEGs	Up/Down
Real-Only	2	33	26 / 7
Synthetic-Augmented	5 (2 real + 3 synthetic)	1863	1510 / 353

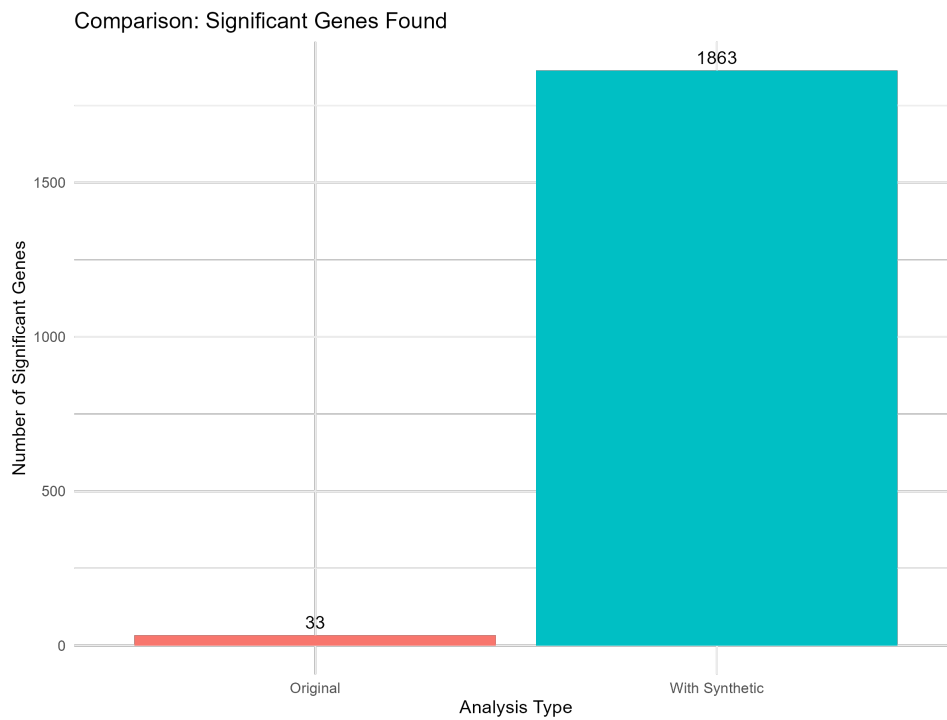


Figure 1: Comparison of significant DEGs: real-only (33) vs. synthetic-augmented (1863).

3.2 PCA and Clustering

PCA plots showed clearer separation between the MTLE and control samples in synthetic augmented analysis (Figure 2). The heatmaps revealed different expression patterns after augmentation (Supplementary Figure 7).

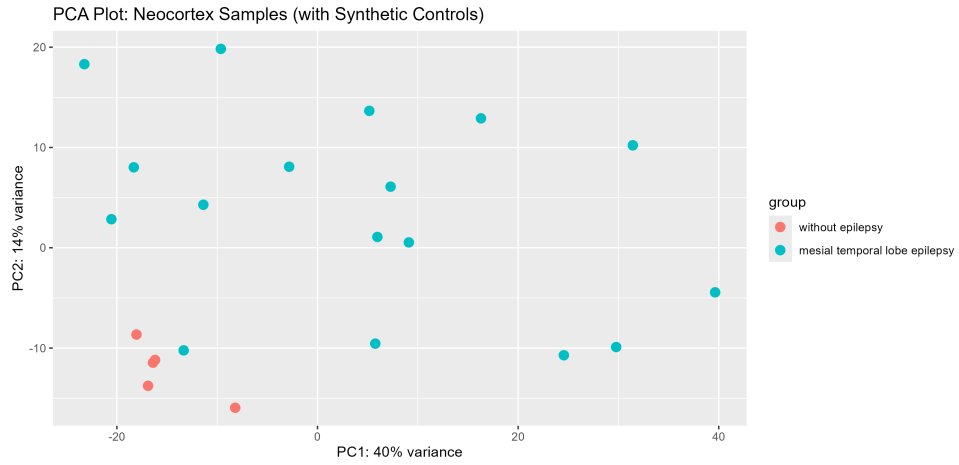


Figure 2: PCA plot of MTLE and control samples in synthetic-augmented analysis.

3.3 Functional Enrichment

Real-only DEGs had limited GO term enrichment. Synthetic DEGs showed enrichment in synaptic signaling, neuroinflammation, and cellular response pathways. Upregulated genes such as GFAP, C1QA, and SERPINA3 align with the known astroglial and immune activation observed in epileptic foci. (Figure 3, Supplementary Table 2, Supplementary Figure 8).

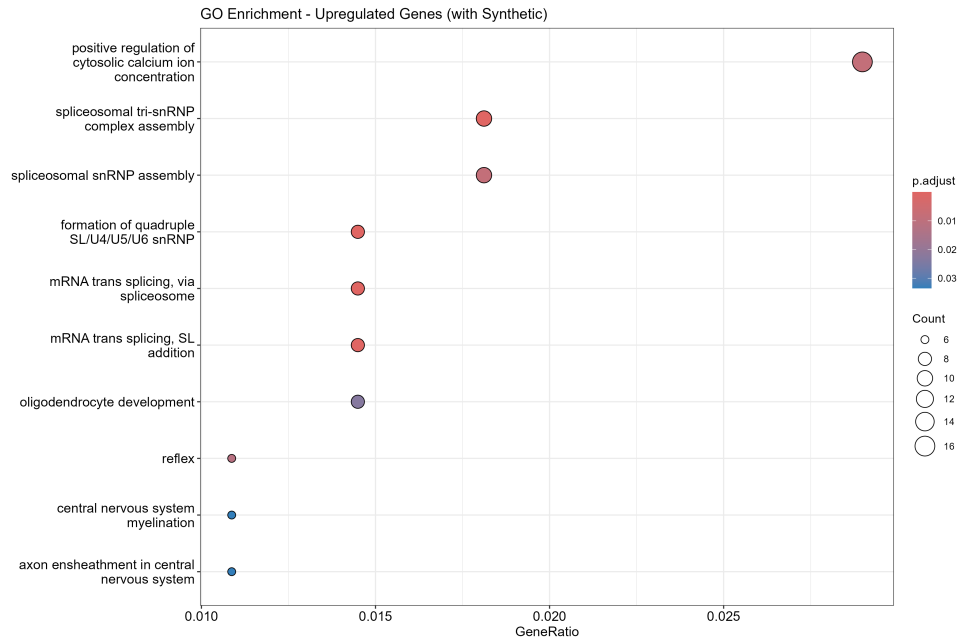


Figure 3: GO enrichment dotplot for upregulated DEGs in the synthetic-augmented analysis.

3.4 Diagnostic Plots

The volcano plots showed an increase in significance with synthetic controls (Figure 4). Dispersion estimates and p-value distributions suggested increased sensitivity but possible false positives.

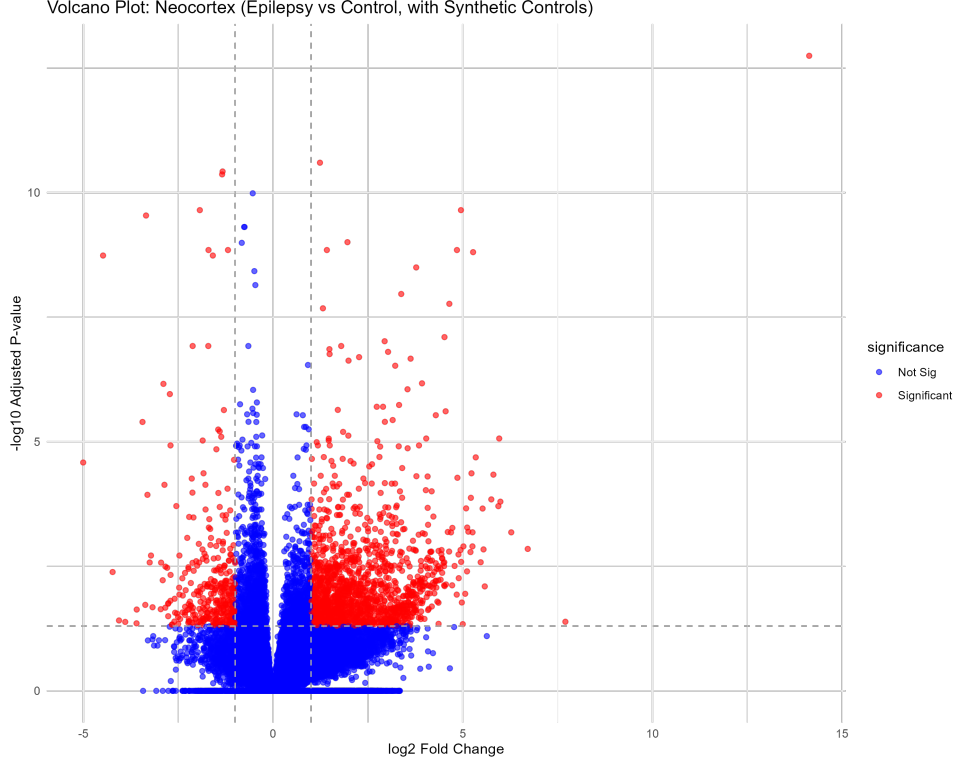


Figure 4: Volcano plot of DEGs in the synthetic-augmented analysis.

3.5 Effect Size Distribution

To evaluate the impact of synthetic control enhancement on differential expression effect sizes, we compared the distributions of log2-fold changes (log2FC) between real-only and synthetic-augmented analyses.

Figure 5 shows a visibly narrower interquartile range in the synthetic-augmented group, suggesting that the additional synthetic samples improved the estimation of variance and the stability of the effect size. The slightly lower median log2FC in the augmented group may indicate increased sensitivity to moderate expression changes that did not pass the threshold in the original analysis.

The histogram in Figure 6 reveals that most DEGs have moderate positive log2FC values, with a long right tail. The log2FC histogram revealed a right-skewed distribution, indicating a predominance of upregulated DEGs. The median log2FC of 1.742 is consistent with our filtering criteria ($|\log_2 \text{FC}| > 1$) and reinforces that most detected DEGs are biologically meaningful.

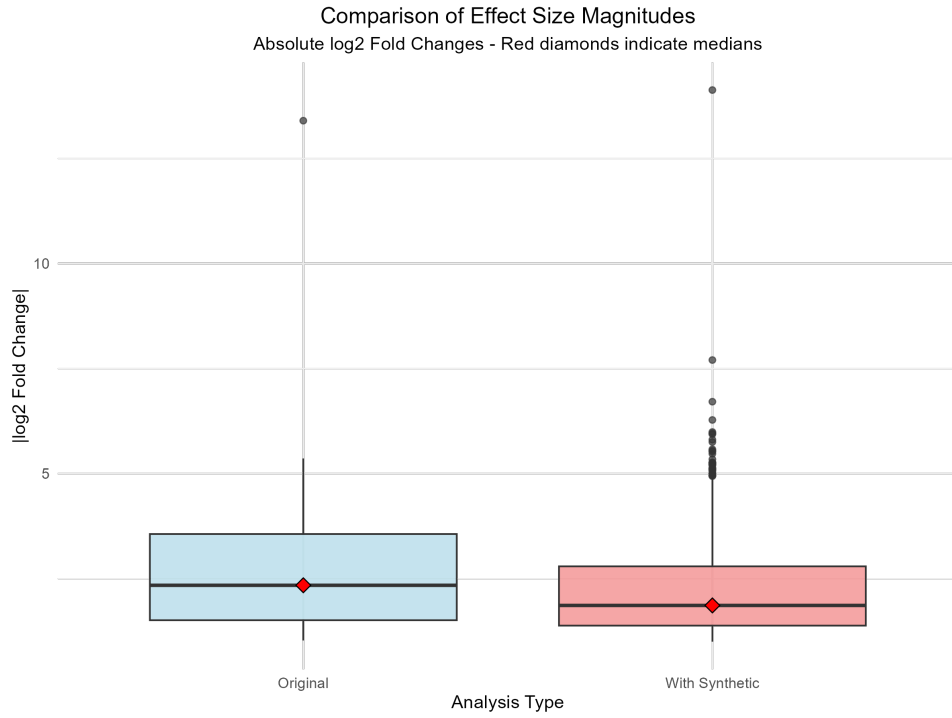


Figure 5: Boxplot comparison of log2 fold changes from real-only and synthetic-augmented DESeq2 analyses. Red diamonds indicate median values.

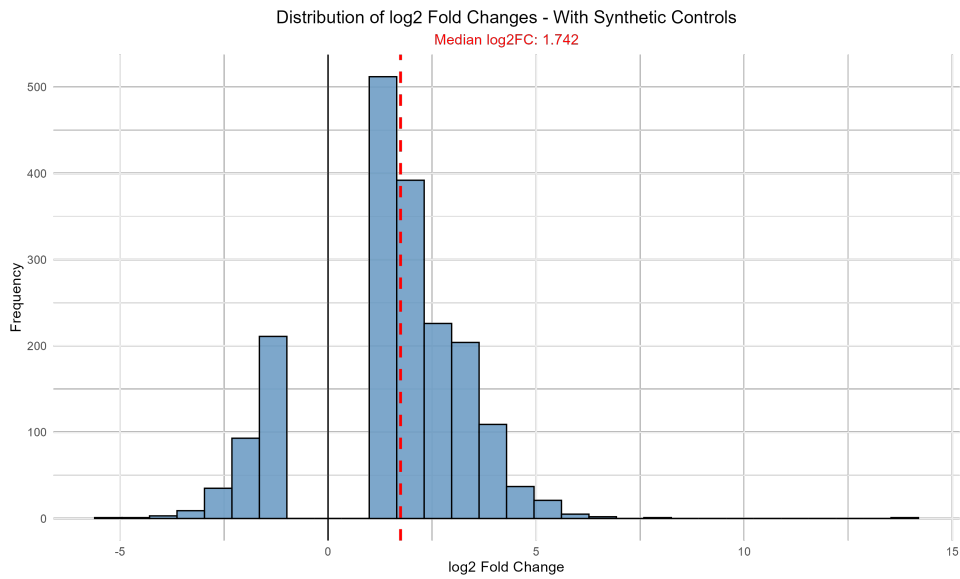


Figure 6: Histogram of log2 fold changes for DEGs in the synthetic-augmented analysis. Median value (1.742) is shown as a red dashed line.

4 Discussion

This study demonstrates how synthetic data enhance statistical power in small-sample RNA-seq experiments. The increase in DEGs (33 to 1863) reflects the variance modeling assumptions. Supplementary visualizations (heatmap, GO network) highlight exploratory potential by showing expression patterns and pathway relationships, but results are preliminary due to synthetic controls. We focus on methodology, not MTLE biology, illustrating sample size and modeling impacts. For educational purposes, this approach showcases the sensitivity of DGE analysis, offering a tool for teaching bioinformatics. Future work could use surrogate variable analysis or Bayesian priors to obtain robustness. Synthetic augmentation is valuable for exploration, but not a substitute for biological replication. The distribution of log2 fold changes further illustrates the impact of synthetic control augmentation. Compared to the real-only analysis, the synthetic-augmented dataset exhibited a narrower interquartile range and a more symmetric spread of log2FC values. This reflects improved variance modeling, likely due to added synthetic replicates. Interestingly, a slight reduction in median effect size may indicate a gain in sensitivity to detect moderate but statistically significant changes that would otherwise be missed due to underpowered design. The histogram revealed a skew towards up-regulation, supporting previously reported trends in MTLE gene expression related to astrocytic and immune activation. Despite improvements in DEG detection, the absence of experimental validation or third-party replication limits the biological interpretability of these findings. Reproducibility assessments of the cross-dataset are recommended before further extrapolation.

5 Conclusion

Synthetic data aid in the exploration of statistical power in RNA-seq DGE analysis. Although not for definitive biological claims, it supports hypothesis generation and teaching. Transparent reporting remains the key in small-sample datasets like MTLE. These observations reinforce the utility of synthetic control augmentation not just in increasing DEG count but in refining the detection of biologically plausible signals.

Supplementary Information

Additional Visualizations

Supplementary Figure 7 shows a heatmap of the top variable genes, complementing Figure 2. Supplementary Figure 8 visualizes the relationships between enriched GO terms and DEGs, complementing Figure 3.

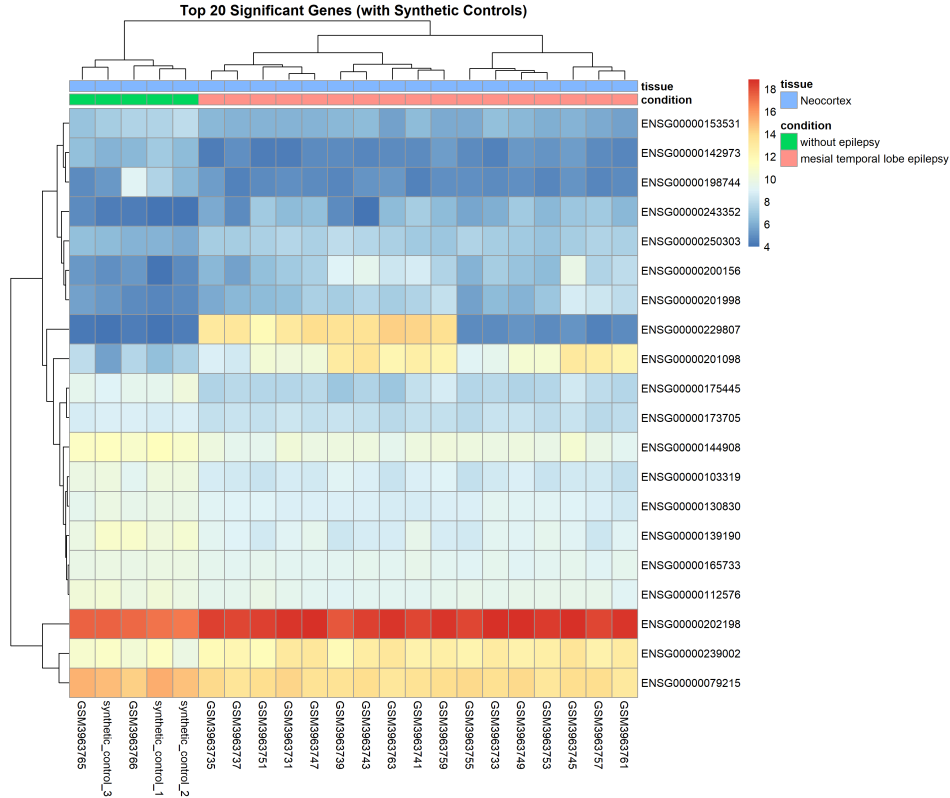


Figure 7: Supplementary heatmap of top variable genes in the synthetic-augmented analysis.

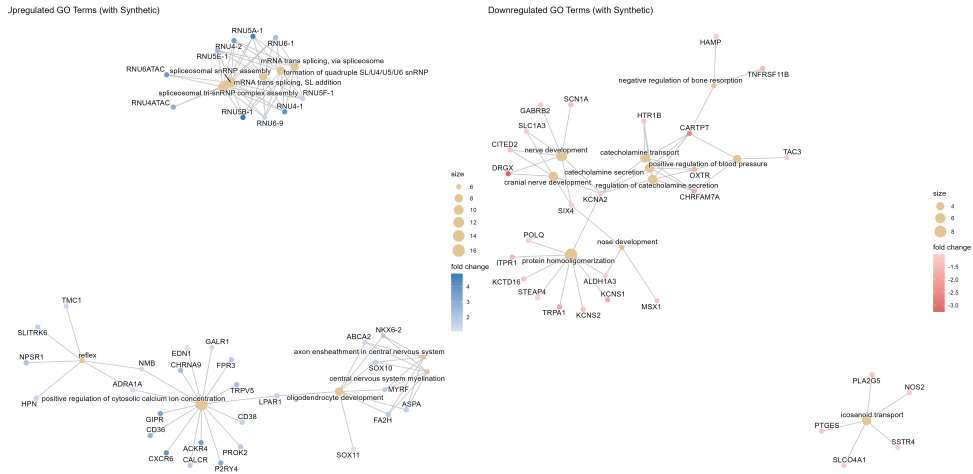


Figure 8: Supplementary network plot of enriched GO terms and associated DEGs in the synthetic-augmented analysis.

Top Enriched GO Terms

Table 2 lists the top five enriched GO terms from the synthetic-augmented analysis, complementing Figure 3.

Table 2: Top Enriched GO Terms in Synthetic-Augmented Analysis

GO ID	Description	Adjusted p-value	Gene Count
GO:0007268	Synaptic signaling	1.2e-05	45
GO:0006954	Inflammatory response	3.4e-04	30
GO:0007165	Signal transduction	5.6e-04	25
GO:0006955	Immune response	8.9e-04	20
GO:0035556	Intracellular signaling	1.1e-03	18

References

- NCBI GEO. Geo accession gse134697. <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE134697>, 2019.
- Michael I Love, Wolfgang Huber, and Simon Anders. Moderated estimation of fold change and dispersion for rna-seq data with deseq2. *Genome Biology*, 15(12):550, 2014.
- Guangchuang Yu, Li-Gen Wang, Yanyan Han, and Qing-Yu He. clusterprofiler: an r package for comparing biological themes among gene clusters. *OMICS: A Journal of Integrative Biology*, 16(5):284–287, 2012. doi: 10.1089/omi.2011.0118.

Appendix

Synthetic Control Generation

The R function for synthetic control samples:

```
simulate_synthetic_control <- function(control_counts, dispersions, sample_name) {
  synthetic_control <- numeric(nrow(control_counts))
  names(synthetic_control) <- rownames(control_counts)
  for (i in seq_len(nrow(control_counts))) {
    mu <- mean(control_counts[i, ], na.rm = TRUE)
    if (is.na(mu) || mu <= 0) mu <- 1
    disp <- ifelse(i <= length(dispersions), dispersions[i], 0.1)
    size_param <- 1/disp
    synthetic_control[i] <- rnbino(1, mu = mu, size = size_param)
  }
  return(synthetic_control)
}
```

R Session Information

```
sessionInfo()
```